Projection–Carry Geometry of Polynomial Number Multiplication

Projection–Carry Geometry Project

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Abstract

We study base-B multiplication through a projection–carry decomposition. Given digit polynomials $X = \sum_i a_i B^i$ and $Y = \sum_j b_j B^j$, their product first forms an outer-product interaction matrix $M_{ij} = a_i b_j$. The raw product coefficients are obtained by diagonal projection,

$$c_k = \sum_{i+j=k} M_{ij},$$

which is equivalently the pushforward of the digit-interaction field under $(i, j) \mapsto i + j$. These raw coefficients are then normalized by the base-B carry recurrence. This separates multiplication into a bilinear projection step and a nonlinear carry-flow step. We formalize carry as a directed mass flow along the degree axis, prove a local carry activation law, and introduce carry-complexity metrics including carry mass, carry count, carry entropy, cascade length, and value-weighted carry. We further connect carry behavior to digit-support geometry: the raw support is the sumset $S_X + S_Y$, and its representation counts bound the possible diagonal mass. Synthetic scaling experiments show that additive energy and maximal representation count are associated with carry mass, while sumset size and support density are associated with carry count and cascade extent. The framework is structural rather than algorithmic, providing a geometric lens on polynomial number multiplication and carry propagation.

1 Introduction

Multiplication is usually treated as scalar arithmetic: two integers enter and one integer exits. In a positional base, however, multiplication has internal geometry. Digits occupy degree positions, digit pairs interact across a two-dimensional index grid, and carry propagation moves excess mass along a one-dimensional degree axis. Projection–Carry Geometry makes this structure explicit.

Fix a base $B \geq 2$. We write nonnegative integers as digit polynomials

$$X = \sum_i a_i B^i, \qquad Y = \sum_j b_j B^j,$$

where $0 \leq a_i, b_j < B$. The product is decomposed into three stages. First, the digit interaction field is the outer-product matrix

$$M_{ij} = a_i b_j.$$

Second, anti-diagonal projection aggregates interactions with the same total degree,

$$c_k = \sum_{i+j=k} M_{ij}.$$

Third, the raw coefficients c_k are converted into valid base-B digits by carry normalization.

The main thesis is that base-B multiplication decomposes into bilinear diagonal projection followed by nonlinear carry flow. The projection step is ordinary convolution of digit polynomials. The carry step is a directed mass flow along the degree axis, locally activated when $c_k + r_k \geq B$, where r_k is incoming carry.

This viewpoint separates potential interaction geometry from realized carry behavior. If $S_X = \{i : a_i \neq 0\}$ and $S_Y = \{j : b_j \neq 0\}$, then the raw support before carry is the sumset $S_X + S_Y$. The representation counts

$$\rho_{X,Y}(k) = \#\{(i,j) \in S_X \times S_Y : i + j = k\}$$

measure concentration of possible interactions on anti-diagonals. We argue that this support-sumset geometry shapes carry complexity, while digit amplitudes and incoming carry determine the realized local events.

The contributions are:

C1. A matrix projection formulation of positional multiplication.

C2. A formulation of carry as nonlinear mass flow.

C3. A local carry activation law.

C4. Support-sumset geometry and representation-count bounds for raw diagonal mass.

C5. Carry-complexity metrics and empirical scaling experiments.

This is a structural and geometric framework, not a new fast multiplication algorithm. Its purpose is to expose the anatomy of multiplication and carry propagation, not to replace existing arithmetic algorithms.

2 Theorem Spine

This section collects the definitions and elementary propositions used throughout the paper.

Digit polynomials. For a base $B \geq 2$, write

$$X = \sum_i a_i B^i, \qquad Y = \sum_j b_j B^j,$$

where $a_i, b_j \in \{0, 1, \ldots, B-1\}$. The digit vectors are indexed from low degree to high degree.

Interaction matrix. The digit interaction matrix is

$$M_{ij} = a_i b_j.$$

It records the pairwise contribution of digit a_i and digit b_j.

Diagonal projection. The raw diagonal coefficients are

$$c_k = \sum_{i+j=k} M_{ij}.$$

Equivalently, diagonal projection pushes the matrix M forward under $(i,j) \mapsto i + j$.

Carry normalization. Set $r_0 = 0$. Carry normalization is defined by

$$s_k = c_k + r_k, \qquad d_k = s_k \bmod B, \qquad r_{k+1} = \left\lfloor \frac{s_k}{B} \right\rfloor.$$

The output digits d_k satisfy $0 \le d_k < B$.

Proposition 1 (Projection equals convolution)**.** *The sequence $c = (c_k)$ is the discrete convolution of the digit sequences $a = (a_i)$ and $b = (b_j)$.*

Proof. For every k,

$$(a * b)_k = \sum_i a_i b_{k-i} = \sum_{i+j=k} a_i b_j = \sum_{i+j=k} M_{ij} = c_k.$$

$\square$

Proposition 2 (Carry preserves value)**.** *If d is obtained from c by base-B carry normalization, then*

$$\sum_k c_k B^k = \sum_k d_k B^k.$$

Proof. The recurrence gives $c_k + r_k = d_k + B r_{k+1}$. Multiplying by B^k and summing over k, the carry terms telescope. Since $r_0 = 0$ and the final carry is exhausted after finitely many positions, the two values are equal. $\square$

Proposition 3 (Local carry activation)**.** *Outgoing carry at degree k is active if and only if*

$$c_k + r_k \ge B.$$

Proof. By definition $r_{k+1} = \lfloor (c_k + r_k)/B \rfloor$. Because $c_k + r_k$ is a nonnegative integer, $r_{k+1} > 0$ exactly when $c_k + r_k \ge B$. $\square$

Digit support and support sumset. Define

$$S_X = \{i : a_i \ne 0\}, \qquad S_Y = \{j : b_j \ne 0\},$$

and

$$S_X + S_Y = \{i + j : i \in S_X,\ j \in S_Y\}.$$

Representation count. For each degree k, define

$$\rho_{X,Y}(k) = \#\{(i,j) \in S_X \times S_Y : i + j = k\}.$$

Proposition 4 (Raw support equals support sumset)**.** *For nonnegative digits,*

$$\operatorname{supp}(c) = S_X + S_Y.$$

Proof. If $k \in S_X + S_Y$, some supported pair (i,j) satisfies $i + j = k$, and then $a_i b_j > 0$, so $c_k > 0$. Conversely, if $c_k > 0$, at least one summand $a_i b_j$ with $i + j = k$ is positive, hence $i \in S_X$ and $j \in S_Y$. $\square$

Proposition 5 (Representation-count bound)**.** *For every degree k,*

$$c_k \le (B-1)^2 \rho_{X,Y}(k).$$

Proof. Each nonzero digit product is at most $(B-1)^2$, and there are $\rho_{X,Y}(k)$ support pairs on the anti-diagonal $i + j = k$. $\square$

Carry complexity metrics. We summarize carry flow by carry mass, carry count, carry depth, carry entropy, maximum cascade length, carry amplification ratio, and value-weighted carry. These metrics quantify the nonlinear normalization cost after bilinear projection.

3 Carry Normalization

Raw coefficients c_k need not be valid base-B digits. Carry normalization defines

$$s_k = c_k + r_k, \qquad d_k = s_k \bmod B, \qquad r_{k+1} = \left\lfloor \frac{s_k}{B} \right\rfloor,$$

with $r_0 = 0$. The normalized result is

$$XY = \sum_k d_k B^k.$$

This step is nonlinear because thresholding and integer division redistribute mass along the degree axis.

3.1 Carry as Nonlinear Mass Flow

After diagonal projection, the product is represented by a sequence of raw coefficients

$$c_k = \sum_{i+j=k} a_i b_j.$$

These coefficients need not be valid base-B digits. We therefore define carry normalization by the recurrence

$$\begin{aligned} r_0 &= 0, \\ s_k &= c_k + r_k, \\ d_k &= s_k \bmod B, \\ r_{k+1} &= \left\lfloor \frac{s_k}{B} \right\rfloor. \end{aligned}$$

Here r_k is the incoming carry at degree k, d_k is the normalized digit kept at degree k, and r_{k+1} is the outgoing carry transported from degree k to degree $k+1$.

This interpretation views carry as a directed mass flow along the degree axis:

$$k \longrightarrow k+1.$$

The mass transported across this edge is r_{k+1}. Thus diagonal projection is a bilinear operation, while carry normalization is a nonlinear flow process.

Proposition 6 (Value preservation of carry normalization)**.** *Let $c = (c_k)$ be a finite sequence of nonnegative integer raw coefficients. Let $d = (d_k)$ be obtained from c by base-B carry normalization. Then*

$$\sum_k c_k B^k = \sum_k d_k B^k.$$

Proof. By the carry recurrence,

$$c_k + r_k = d_k + Br_{k+1}.$$

Multiplying by B^k, we obtain

$$c_k B^k + r_k B^k = d_k B^k + r_{k+1} B^{k+1}.$$

Summing over k, the carry terms telescope. Since $r_0 = 0$ and the final carry is eventually exhausted, we get

$$\sum_k c_k B^k = \sum_k d_k B^k.$$

□

4 Local Carry Laws

Carry normalization is governed by a local threshold rule. At degree k, let

$$s_k = c_k + r_k,$$

where c_k is the raw coefficient and r_k is the incoming carry. Then

$$d_k = s_k \bmod B, \qquad r_{k+1} = \left\lfloor \frac{s_k}{B} \right\rfloor.$$

Proposition 7 (Local Carry Activation)**.** *The outgoing carry at degree k is active if and only if*

$$c_k + r_k \geq B.$$

Proof. By definition,

$$r_{k+1} = \left\lfloor \frac{c_k + r_k}{B} \right\rfloor.$$

Since $c_k + r_k$ is a nonnegative integer and $B \geq 2$, the floor is positive exactly when $c_k + r_k \geq B$. Thus $r_{k+1} > 0$ if and only if $c_k + r_k \geq B$. □

Carry cascade length. A carry cascade is a maximal consecutive block of active carry edges

$$k \to k + 1.$$

The cascade length is the number of active edges in such a block. The maximum carry cascade length records the longest uninterrupted carry propagation event.

Value-weighted carry flow. The value-weighted carry flow is

$$\mathrm{VWC}(c) = \sum_k r_{k+1} B^{k+1}.$$

It measures transported carry mass in positional value units rather than raw edge-count units. A normalized version divides this quantity by $\max(\sum_k c_k B^k, 1)$.

Carry is local in recurrence but global in consequence because outgoing carry changes the next degree. Thus c_k alone does not determine activation: the correct local state variable is $s_k = c_k + r_k$.

5 Support-Sumset Geometry

Digit support. For a digit vector $a = (a_i)$, its support is

$$S_X = \{i : a_i \neq 0\}.$$

Similarly $S_Y = \{j : b_j \neq 0\}$.

Support sumset. The support sumset is

$$S_X + S_Y = \{i + j : i \in S_X,\ j \in S_Y\}.$$

It is the degree-level footprint of all possible digit interactions before carry normalization.

Representation counts. For each degree k, define

$$\rho_{X,Y}(k) = \#\{(i,j) \in S_X \times S_Y : i + j = k\}.$$

The number $\rho_{X,Y}(k)$ is the diagonal density of the support geometry. The additive energy is

$$E(S_X, S_Y) = \sum_k \rho_{X,Y}(k)^2,$$

which measures concentration of representation counts across anti-diagonals.

Proposition 8 (Raw support as a sumset)**.** *Assume the digits are nonnegative. Let*

$$c_k = \sum_{i+j=k} a_i b_j.$$

Then

$$\operatorname{supp}(c) = S_X + S_Y.$$

Proof. If $k \in S_X + S_Y$, then there exist $i \in S_X$ and $j \in S_Y$ with $i + j = k$. Since digits are nonnegative and supported digits are positive, $a_i b_j > 0$, so $c_k > 0$. Conversely, if $c_k > 0$, at least one summand $a_i b_j$ with $i + j = k$ is positive. Hence $i \in S_X$, $j \in S_Y$, and $k \in S_X + S_Y$. $\square$

Proposition 9 (Representation-count bound)**.** *For base B, every raw coefficient satisfies*

$$c_k \leq (B-1)^2 \rho_{X,Y}(k).$$

Proof. Every digit product satisfies $a_i b_j \leq (B-1)^2$. At degree k, there are exactly ρ_k support pairs that can contribute nonzero products. Summing these products gives the stated bound. $\square$

Carry complexity is influenced by concentration of $\rho_{X,Y}(k)$. Large representation counts and high additive energy indicate many support-pair collisions on the same anti-diagonals, increasing the potential raw mass that carry normalization must absorb. The realized carry still depends on digit values and incoming carry, but support-sumset geometry provides a structural predictor for where nonlinear normalization may become active.

6 Empirical Scaling Experiments

Experimental setup. The synthetic scaling experiments sample random digit supports and digit amplitudes across bases $B \in \{2, 10, 16\}$, digit lengths such as 8 and 16 in the default reproducible grid, and support densities $\{0.2, 0.5, 0.7, 1.0\}$. For each configuration, random pairs are generated and passed through the projection–carry pipeline. The generated summary and correlation tables are stored in the `paper/tables` directory.

Predictors and outcomes. The geometry predictors are sumset size, maximum representation count, additive energy, support densities, and diagonal mass concentration. The carry outcomes are carry mass, carry count, maximum carry cascade length, carry amplification ratio, and normalized value-weighted carry. We also examine normalized quantities such as carry mass per digit and cascade fraction.

Raw support concentration and carry mass. Figure 1 shows additive energy against carry mass. Additive energy measures concentration of representation counts on anti-diagonals; high energy indicates that many support pairs may collide on a small set of degrees. Figure 2 gives the related view using the largest representation count. In the synthetic grid, both quantities behave as structural predictors for absolute carry mass.

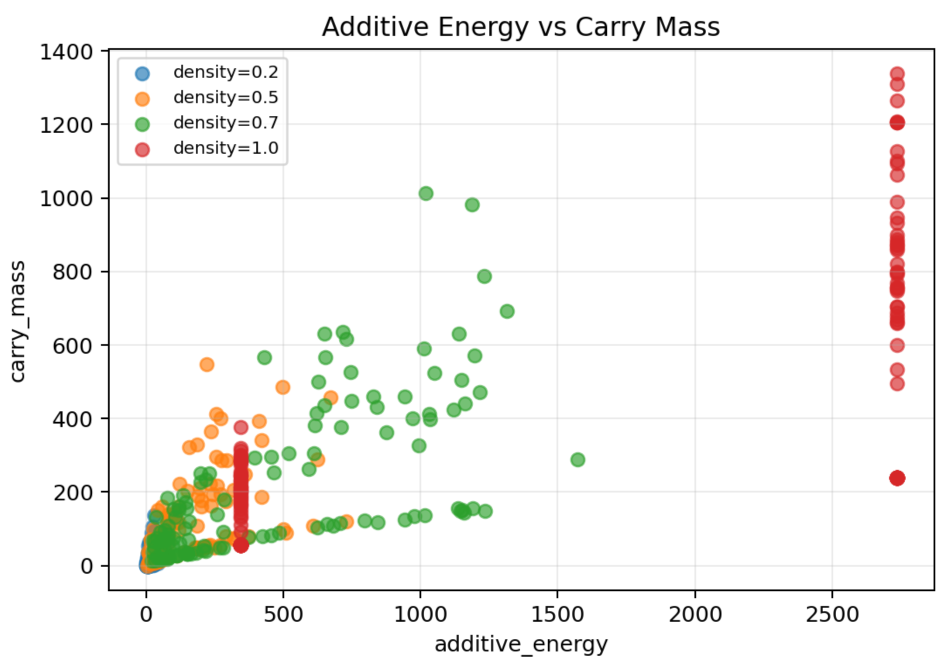

Figure 1: Additive energy versus carry mass.

Spatial extent of carry flow. Figure 3 compares sumset size with carry count. Sumset size tracks how many degree positions can receive raw mass; this is therefore related to the spatial extent of carry activity. Figure 4 shows that support density is associated with cascade fraction, the normalized length of the longest consecutive active-carry segment.

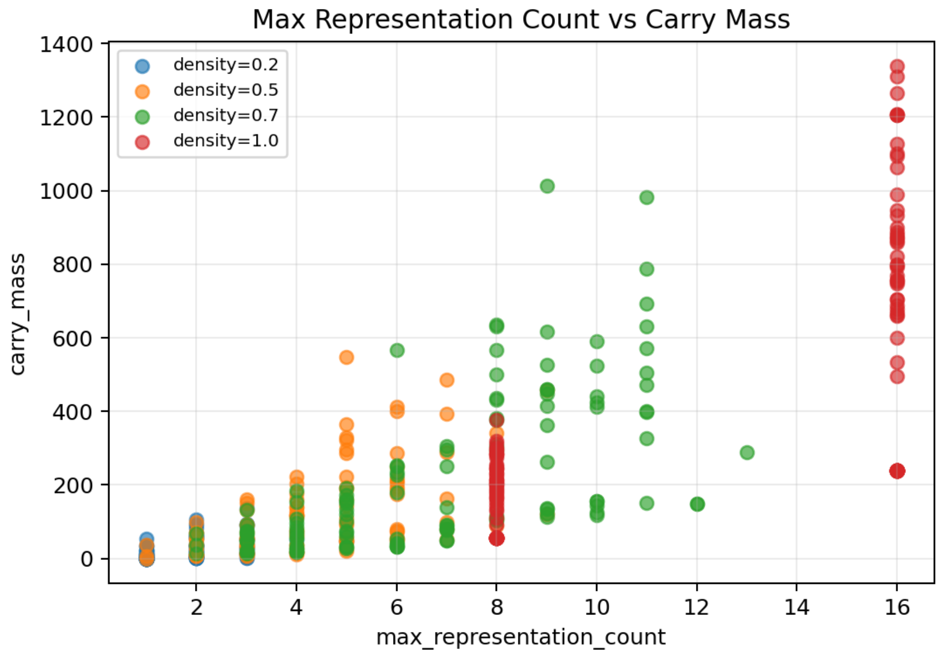

Figure 2: Maximum representation count versus carry mass.

Base effects and normalized metrics. Base affects absolute carry mass and normalized amplification differently. Figure 5 shows carry amplification by base, while Figure 6 shows carry mass per digit. Larger bases raise the local activation threshold but also allow larger digit products. The normalized plots should therefore be read together with the raw plots rather than as replacements.

Interpreting normalization. Figure 7 shows normalized additive energy against carry mass per digit. Normalized additive energy alone is insufficient because it removes scale: two supports can have similar normalized concentration but very different total interaction mass. This reinforces the central interpretation: support geometry gives potential concentration, while digit amplitudes and incoming carry determine realized carry flow.

7 Discussion and Limitations

Projection–Carry Geometry is not a new multiplication algorithm. It is a structural and geometric framework for the ordinary base-B multiplication process. The goal is to separate bilinear projection from nonlinear carry flow and to make the geometry of carry propagation measurable.

Support geometry controls potential mass concentration, not exact carry. The sumset $S_X + S_Y$, representation counts $\rho_{X,Y}(k)$, and additive energy describe where digit interactions can collide. They do not determine the actual raw coefficient values. Digit amplitudes determine c_k, and incoming carry determines the realized activation condition $c_k + r_k \geq B$. Thus carry is locally activated but globally propagated through the recurrence.

The experiments are synthetic random digit experiments. They are useful for testing structural hypotheses and producing reproducible scaling tables, but they do not characterize every input distribution that appears in arithmetic software or hardware. Deterministic support families may display different behavior.

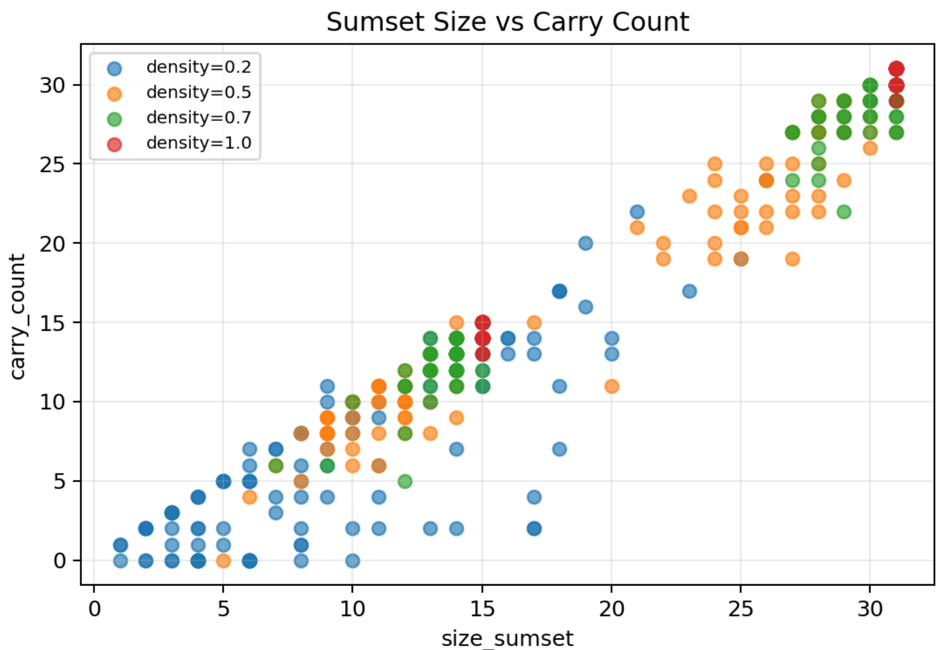

Figure 3: Support-sumset size versus carry count.

Normalized metrics must also be interpreted carefully. Normalization by digit length, support size, or total interaction count can remove the scale that drives absolute carry mass. For this reason, normalized figures should be read alongside raw carry and raw support-concentration plots.

Future work includes binary multiplication and bit-level circuits, multi-factor tensor projection, signed digits, negative coefficients, carry minimization, and more direct connections to additive combinatorics.

8 Conclusion

Base-B multiplication can be viewed as bilinear projection followed by nonlinear carry flow. The outer-product matrix records pairwise digit interactions, anti-diagonal projection aggregates those interactions into raw degree coefficients, and carry normalization transports excess mass along the degree axis while preserving numerical value.

Carry is locally activated, globally propagated, and geometrically conditioned. It is locally activated by the threshold $c_k + r_k \geq B$, globally propagated because outgoing carry changes the next degree, and geometrically conditioned because support-sumset representation counts control where raw mass can concentrate. This gives a coherent language for studying multiplication as projection followed by normalization, and for relating carry complexity to the geometry of digit supports.

References

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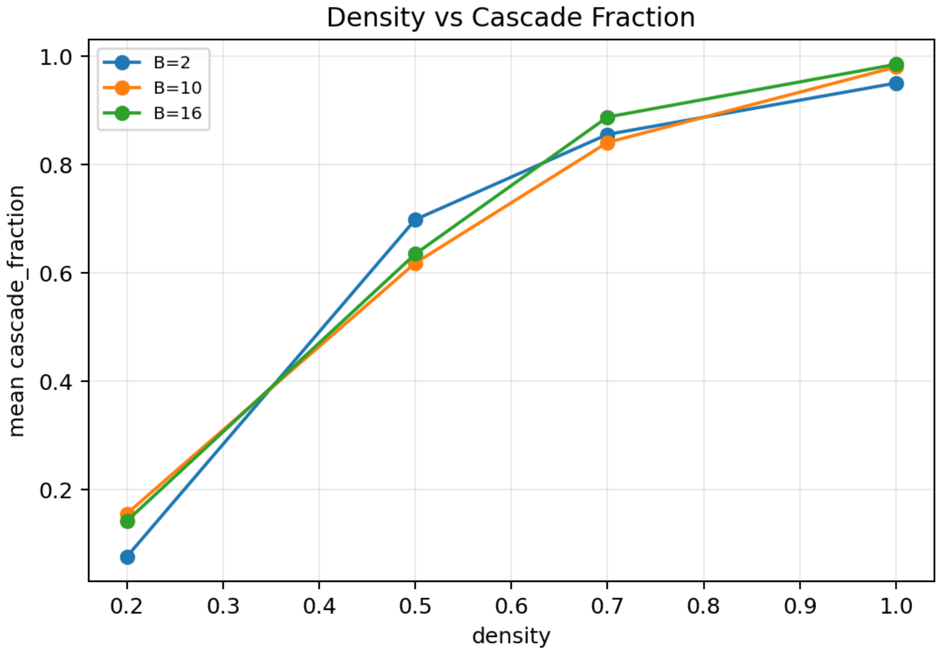

Figure 4: Support density versus cascade fraction.

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[3] Terence Tao and Van H. Vu. *Additive Combinatorics*. Cambridge University Press, 2006.

[4] Joachim von zur Gathen and Jürgen Gerhard. *Modern Computer Algebra*. Cambridge University Press, 2013.

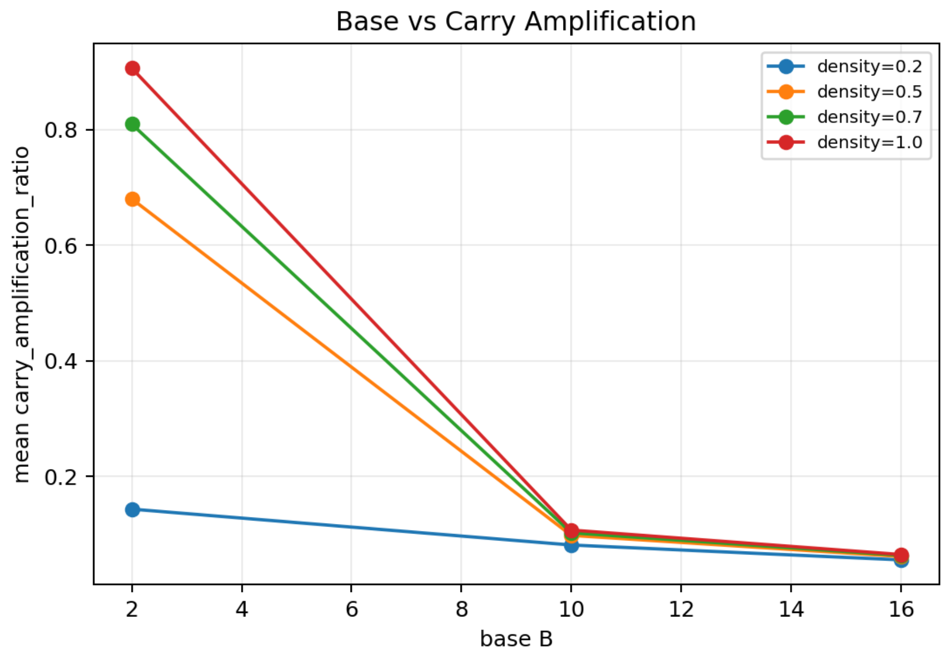

Figure 5: Base versus carry amplification ratio.

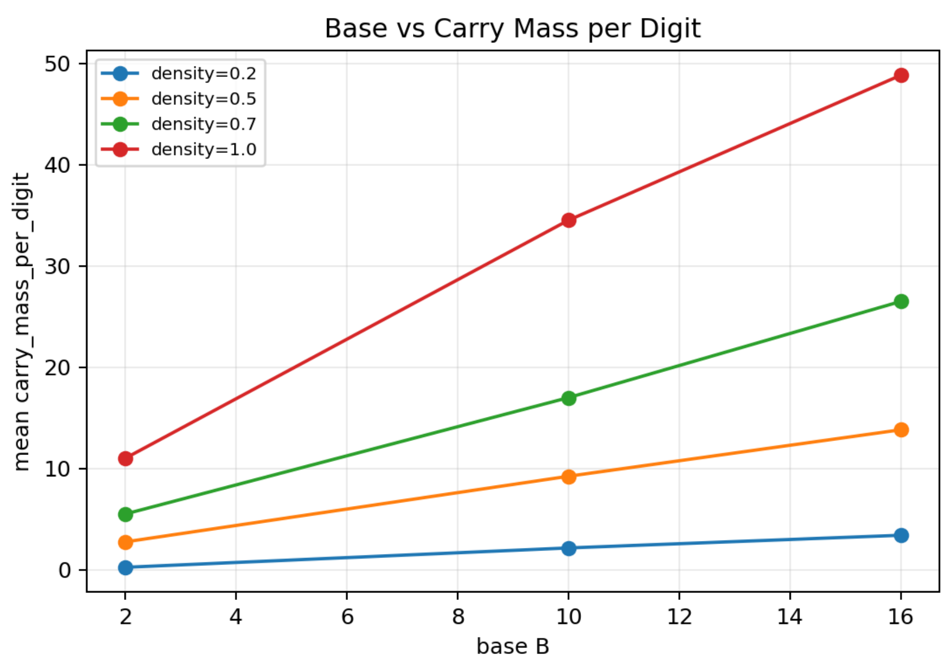

Figure 6: Base versus carry mass per digit.

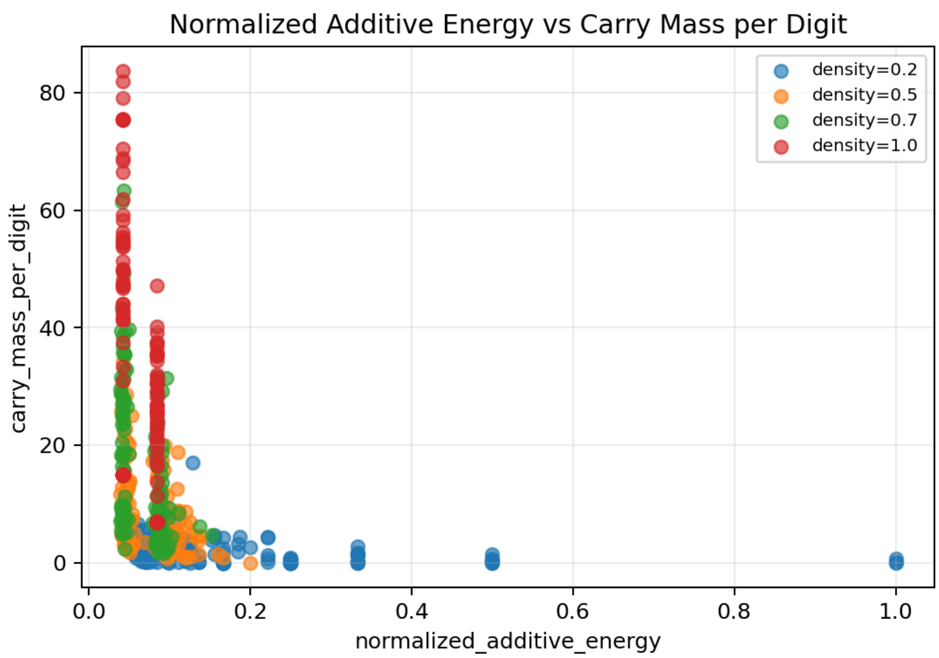

Figure 7: Normalized additive energy versus carry mass per digit.